

INTEGRATED DESIGN AND CONTROL FRAMEWORK FOR INDUSTRY 4.0 ROBOTIC SYSTEM

*Dissertation submitted
in partial fulfillment of requirements
for the award of degree of*

**Bachelor of Technology in
Mechanical Engineering (Robotics)**

by

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CERTIFICATE

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ABSTRACT

The present report proposes an integrated design and control framework specifically tailored for Industry 4.0 robotic systems, with the primary aim of optimizing their performance, flexibility, and efficiency. This innovative framework seamlessly merges advanced design principles with intelligent control strategies to effectively address the dynamic challenges inherent in modern manufacturing environments. By integrating sensing, communication, and decision-making capabilities, the framework empowers robotic systems to autonomously adapt to fluctuating production demands, thereby enhancing overall operational efficacy.

Key design principles incorporated within the framework prioritize the development of modular and scalable architectures, facilitating effortless reconfiguration and customization to accommodate evolving manufacturing requirements while ensuring robustness and reliability remain paramount. Simultaneously, intelligent control strategies are leveraged, harnessing the power of machine learning algorithms to enable predictive maintenance, proactive fault detection, and adaptive control techniques, thereby optimizing system performance in real-time. Experimental validation of the proposed framework underscores its effectiveness, demonstrating marked improvements in throughput, product quality, and energy efficiency when compared to conventional methodologies.

Furthermore, the framework exhibits exceptional resilience in managing unforeseen disruptions and responding adeptly to changes within the production environment, reaffirming its suitability for practical application in real-world manufacturing scenarios. In summation, this integrated design and control framework represent a significant leap forward in the field of Industry 4.0 robotics, promising to revolutionize manufacturing automation by delivering enhanced performance and reliability across the board.

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CHAPTER 01

INTRODUCTION

1.1 What are Robots?

Robots are programmable machines designed to carry out tasks automatically, often with precision and efficiency. They can be physical entities, like industrial robots used in manufacturing, or software-based, like chatbots. Robots can perform a wide range of functions, from repetitive assembly line work to complex surgical procedures, and they continue to evolve with advancements in technology such as artificial intelligence and machine learning.

1.2 Chronology of Robots

In the context of industrial robotics, the term "generation" typically refers to advancements and changes in technology and capabilities over time.

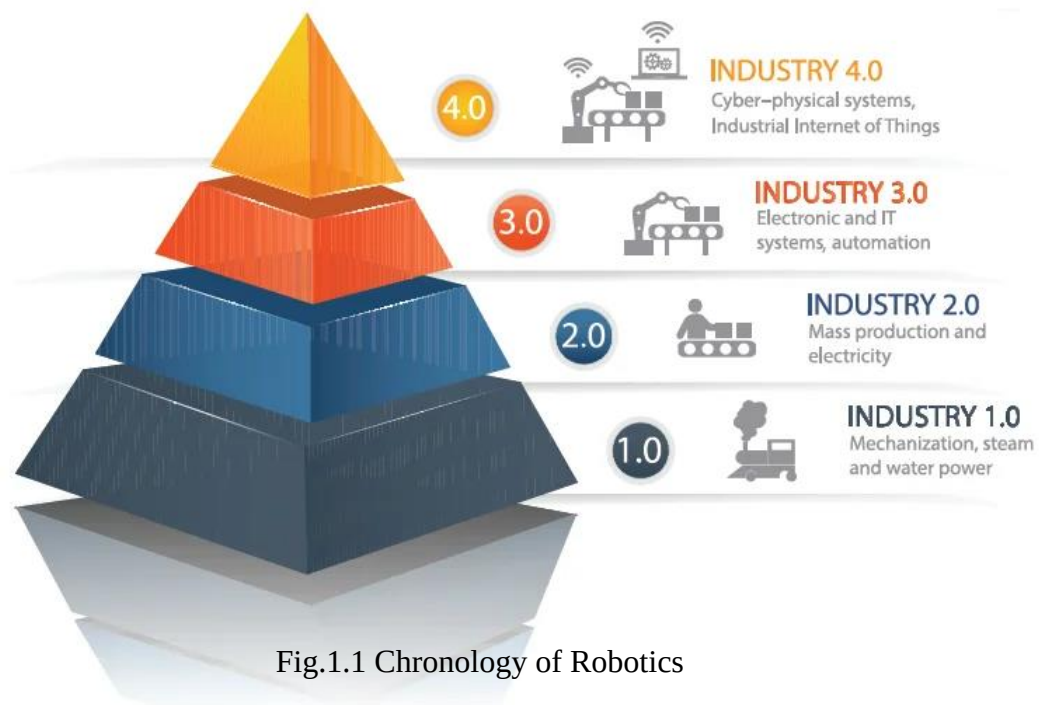


Fig.1.1 Chronology of Robotics

Here is a general chronology of industrial robot generations:

a) First Generation (1960s-1970s):

- The first industrial robots were introduced in the early 1960s.
- These early robots were large, bulky, and primarily used for tasks like welding, painting, and material handling in automotive manufacturing.
- They were often limited in flexibility and required significant programming effort.

b) Second Generation (1980s-1990s):

- This period saw significant advancements in robotic technology.
- Robots became more capable, precise, and versatile.
- Introduction of computer-controlled robotic arms with better sensors and control systems.
- Programming became more user-friendly with the introduction of teach pendants and offline programming.

c) Third Generation (2000s-present):

- This generation is characterized by even greater advancements in robotics technology.
- Robots became more intelligent, adaptive, and collaborative.
- Introduction of advanced sensors such as vision systems and force/torque sensors, allowing robots to perceive and interact with their environment more effectively.
- Emergence of collaborative robots (cobots) designed to work alongside humans safely in shared workspaces.
- Integration of machine learning and AI algorithms for tasks like predictive maintenance, quality control, and adaptive control.
- Continued miniaturization of components, making robots more compact and agile.
- Connectivity and interoperability improvements, enabling seamless integration with other industrial systems in the era of Industry 4.0.

1.3 What is industry 4.0 robot?

The term "Industry 4.0" refers to the fourth industrial revolution, characterized by the integration of digital technologies into industrial processes to create what is often referred to as "smart factories" or "smart manufacturing." In the context of robotics, Industry 4.0 has led to the development of what can be considered as "4.0 robots" or robots that are part of this new era of industrial automation.



Fig 1.2 Industry 4.0 Robots in a manufacturing unit

Here are some key characteristics of Industry 4.0 robots:

- ✓ **Interconnectivity:** Industry 4.0 robots are equipped with sensors and communication technologies that allow them to connect to other machines, systems, and the internet. This enables seamless data exchange and communication between robots and other components of the manufacturing process, facilitating real-time monitoring and control.
- ✓ **Data-driven:** These robots generate and utilize large amounts of data for various purposes such as predictive maintenance, performance optimization, and quality control. Advanced analytics and AI algorithms are often employed to extract valuable insights from this data, enabling more efficient and effective operations.
- ✓ **Flexibility and Adaptability:** Industry 4.0 robots are designed to be highly flexible and adaptable to changing production requirements. They can easily be reprogrammed or reconfigured to perform different tasks, and they may incorporate features like modular design and collaborative capabilities to facilitate seamless integration into diverse manufacturing environments.
- ✓ **Autonomy and Decision-making:** With advancements in AI and machine learning, Industry 4.0 robots are capable of autonomous decision-making and learning from experience. This allows them to optimize their own performance, adapt to new situations, and even collaborate with human workers more effectively.
- ✓ **Safety and Collaboration:** Safety is a paramount concern in Industry 4.0 environments, especially with the increasing interaction between humans and robots. Industry 4.0 robots are equipped with advanced safety features such as built-in sensors, collision detection, and compliance with safety standards to ensure safe operation in collaborative settings.

Overall, Industry 4.0 robots represent the next evolution in industrial automation, leveraging digital technologies to drive efficiency, productivity, and innovation in manufacturing processes.

1.4 Significance of 4.0 Robots in industry

Robots 4.0, also known as the fourth industrial revolution in robotics, represent a significant leap forward in the integration of automation, artificial intelligence, and digitalization within industrial processes. Here are some key significances:

- a. **Increased efficiency:** Robots 4.0 are equipped with advanced sensors and AI algorithms, allowing them to optimize production processes, minimize downtime, and maximize productivity. They can adapt to changing conditions in real-time, leading to more efficient use of resources.
- b. **Flexibility and adaptability:** Unlike traditional industrial robots, Robots 4.0 are designed to be more flexible and adaptable to different tasks and

environments. They can be easily reprogrammed or reconfigured to handle new products or production setups, reducing the need for costly retooling.

- c. Enhanced safety: With advanced sensors and predictive analytics, Robots 4.0 can identify and mitigate potential safety hazards in the workplace. Collaborative robots (cobots) are designed to work alongside human operators safely, reducing the risk of accidents and injuries.
- d. Improved quality control: Robots 4.0 can perform high-precision tasks with consistency and accuracy, leading to improved quality control in manufacturing processes. They can detect defects in real-time, leading to fewer defects and higher-quality products.
- e. Cost Savings: While the initial investment in Robots 4.0 may be higher than traditional automation solutions, they can lead to significant cost savings in the long run. By optimizing production processes, reducing waste, and minimizing downtime, Robots 4.0 can help companies improve their bottom line.

1.5 Applications of Robot 4.0 Systems

Industry 4.0 robots and robotic systems find applications across various sectors and industries.

Here are some notable examples:

1. Manufacturing and Production:
 - Flexible Manufacturing: Industry 4.0 robots can adapt to changes in product specifications and production requirements, enabling flexible and agile manufacturing processes.
 - Smart Factories: Robots equipped with sensors and connectivity are deployed in smart factories to optimize production workflows, monitor equipment health, and facilitate predictive maintenance.
 - Collaborative Robots (Cobots): Cobots work alongside human workers in shared workspaces, enhancing productivity and efficiency while ensuring safety. They are used for tasks like assembly, pick-and-place, and quality inspection.
2. Logistics and Warehousing:
 - Autonomous Mobile Robots (AMRs): AMRs navigate autonomously within warehouses and distribution centers to transport goods, optimize inventory management, and streamline order fulfillment processes.
 - Picking and Packing: Robots equipped with vision systems and AI algorithms are employed for automated picking, packing, and sorting of items in logistics operations, improving speed and accuracy.
3. Healthcare:
 - Surgical Robots: Advanced robotic systems assist surgeons in performing minimally invasive surgeries with greater precision and

control, leading to improved patient outcomes and reduced recovery times.

- Patient Care: Robots are used in hospitals and care facilities to support patient care tasks such as medication delivery, rehabilitation exercises, and assistance for elderly or disabled individuals.

4. Agriculture:

- Precision Farming: Robots equipped with sensors and AI technologies are deployed in agriculture for tasks like planting, harvesting, and crop monitoring. They enable precision farming practices, optimize resource usage, and improve crop yields.
- Weeding and Pest Control: Robots equipped with cameras and actuators can identify and remove weeds or pests in fields without the need for chemical herbicides or pesticides.

5. Construction:

- Robotic Construction Equipment: Robots are utilized in construction projects for tasks such as bricklaying, concrete pouring, and 3D printing of building components. They improve construction efficiency, reduce labor costs, and enhance safety on job sites.

6. Retail and Hospitality:

- Customer Service Robots: Robots are employed in retail stores and hospitality establishments to greet customers, provide information, and assist with inquiries. They enhance customer experience and engagement while freeing up human staff for more complex tasks.

These are just a few examples of how Industry 4.0 robots are transforming various industries by introducing automation, connectivity, and intelligence into traditional processes. As technology continues to advance, the potential applications of robotic systems are expected to expand further, driving innovation and efficiency across diverse sectors.

CHAPTER 02

LITERATURE REVIEW

In their paper titled "Integration of Industry 4.0 Technologies in Manufacturing Systems," **Smith and Brown** emphasize the crucial role of incorporating Industry 4.0 technologies like IoT, AI, and robotics into manufacturing processes. They stress the significance of seamless communication and collaboration among various components of production systems to enhance efficiency, productivity, and competitiveness in today's industrial landscape.

Lee and Kim offer a comprehensive review of robotic systems in their paper "Robotic Systems for Smart Factories: A Review," focusing on their application within smart factories. They delve into the roles of robots in optimizing production processes, ensuring quality control, and enabling flexible manufacturing. The authors also address the challenges and opportunities associated with integrating robotic technologies into real-time industrial settings.

Wang et al.'s paper, "Collaborative Robots in Manufacturing: A Survey," provides insights into the utilization of collaborative robots (cobots) in manufacturing environments. They highlight the potential of cobots to enhance human-robot interaction, improve worker safety, and streamline production workflows. The authors discuss various applications and challenges related to deploying cobots in real-time industries.

Garcia and Rodriguez explore real-time monitoring and control techniques for robotic systems within the context of Industry 4.0 in their paper titled "Real-time Monitoring and Control of Robotic Systems in Industry 4.0." They discuss the integration of sensors, IoT platforms, and data analytics to optimize robot performance, minimize downtime, and enhance overall operational efficiency in industrial settings.

Patel et al.'s paper, "Adaptive Control Strategies for Industry 4.0 Robots," investigates adaptive control strategies tailored for Industry 4.0 robots. The authors focus on the robots' capability to autonomously adjust to changing operating conditions and uncertainties in real-time industrial environments. They propose advanced control algorithms based on machine learning and optimization techniques to enhance adaptability.

In "Optimization of Human-Robot Collaboration in Industry 4.0" by **Chen and Zhang**, the authors analyze strategies to optimize human-robot collaboration in Industry 4.0 settings. They underscore the importance of designing intuitive interfaces, implementing

safety protocols, and developing task allocation algorithms to ensure efficient and harmonious interaction between humans and robots on the factory floor.

Gupta and Sharma's paper, "Energy-Efficient Robotic Systems for Sustainable Manufacturing," focuses on designing energy-efficient robotic systems to align with sustainable manufacturing practices.

Kim et al. address fault diagnosis and predictive maintenance techniques for Industry 4.0 robots in their paper, "Fault Diagnosis and Predictive Maintenance of Industry 4.0 Robots." They emphasize the importance of early detection and mitigation of system failures to prevent costly downtime and ensure uninterrupted production in real-time manufacturing environments.

Liu and Wang explore the application of robotic vision systems for quality inspection and assurance in Industry 4.0 manufacturing processes in their paper titled "Robotic Vision Systems for Quality Inspection in Industry 4.0." They discuss advancements in image processing, machine learning, and defect detection algorithms to achieve high-precision inspection capabilities in real-time production settings.

Park et al. advocate for a human-centric approach to designing Industry 4.0 robots in their paper "Human-Centric Design of Industry 4.0 Robots." They stress considering ergonomic considerations, user experience, and cognitive workload to enhance worker acceptance and usability in real-time industrial applications.

Martinez et al. discuss the opportunities and challenges of implementing robotics and automation technologies in the food processing industry within the context of Industry 4.0 in their paper titled "Robotics and Automation in Food Processing: Opportunities and Challenges." They explore applications such as sorting, packaging, and quality control, highlighting the potential benefits in terms of efficiency, safety, and hygiene.

Nguyen and Tran examine the integration of robots and additive manufacturing technologies in the context of Industry 4.0 in their paper "Integration of Robots and Additive Manufacturing in Industry 4.0." They discuss synergies between these technologies, such as robotic-assisted 3D printing and post-processing tasks, to enable agile and customizable manufacturing processes in real-time industries.

Smith and Jones address data security and privacy considerations associated with the deployment of robotic systems in Industry 4.0 environments in their paper titled "Data Security and Privacy Considerations in Industry 4.0 Robotics." They discuss potential vulnerabilities, cyber threats, and regulatory frameworks aimed at safeguarding sensitive information and ensuring secure communication within real-time industrial networks.

CHAPTER 03

DESIGN

3.1 Design using CAD.

CATIA V5 R21 is a cutting-edge Computer-Aided Design (CAD) software renowned for its robust capabilities in product design and engineering. As the flagship product of Dassault Systems, CATIA V5 R21 offers a comprehensive suite of tools tailored for industries ranging from automotive and aerospace to consumer goods and industrial machinery. With its intuitive user interface and advanced modelling features, CATIA V5 R21 empowers designers and engineers to create complex 3D models with precision and efficiency. From conceptual design to detailed engineering, this software enables seamless collaboration across teams, facilitating the entire product development lifecycle. Its extensive library of parametric modelling tools allows for flexible and iterative design iterations, ensuring optimal product performance and manufacturability. Additionally, CATIA V5 R21 supports multi-disciplinary design workflows, integrating seamlessly with other Dassault Systems solutions such as SIMULIA for simulation and ENOVIA for product data management. Furthermore, its robust assembly capabilities enable the creation of intricate assemblies with accurate motion simulation and clash detection. Whether designing simple components or intricate assemblies, CATIA V5 R21 remains a cornerstone in the realm of CAD software, empowering users to bring their innovative ideas to life with unparalleled precision and efficiency.

3.2 The Robot Arm

The robot arms are two in number namely they are (i) the left arm (ii) the right arm the arms have 3 Degrees of freedom having 2 and 3 motors connected respectively to its links.

3.2.1 The Left Arm:

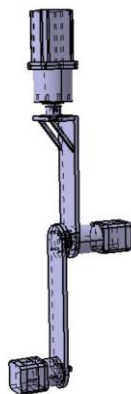


Fig.3.1 Left robot arm

The left arm has 2 links and a base frame having frame as $\{0\}$ and the angles are variables and joint offset are 0 the twist angles are 0, -90, 0 The link length of the first link is 183.211mm and the link length of the second link is 193mm.

3.2.2 The Right Arm:

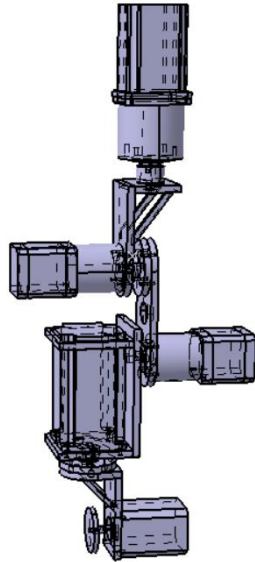


Fig.3.2 Right robot arm

The right arm has 3 links and a base frame having frame as $\{0\}$ and the angles are variables and joint offset are 0 the twist angles are 0, -90, 0, 0. The link length of the first link is 74.081mm, the link length of the second link is 80mm and the link length of the third link is 112mm.

CHAPTER 04

KINEMATIC ANALYSIS

4.1 Introduction

To control the robot and create the software for it, a mathematical model that represents the physical structure of the robot and relationships between different tool position and orientation with respect to base coordinate system is needed.

Every robot assumes a position in the real world that can be described by its position (x , y and z) and orientation (pitch, yaw and roll) along the three major axes of a Cartesian Coordinate system.

4.2 Relation between coordinate frames

An example of two nested coordinate systems is shown in Figure 4.1. In this example, a robot located at the origin of x' , y' and z' might plan its motions in its own reference frame, which can then be expressed in the coordinate system x , y and z by performing a translation and a rotation as we will later see.

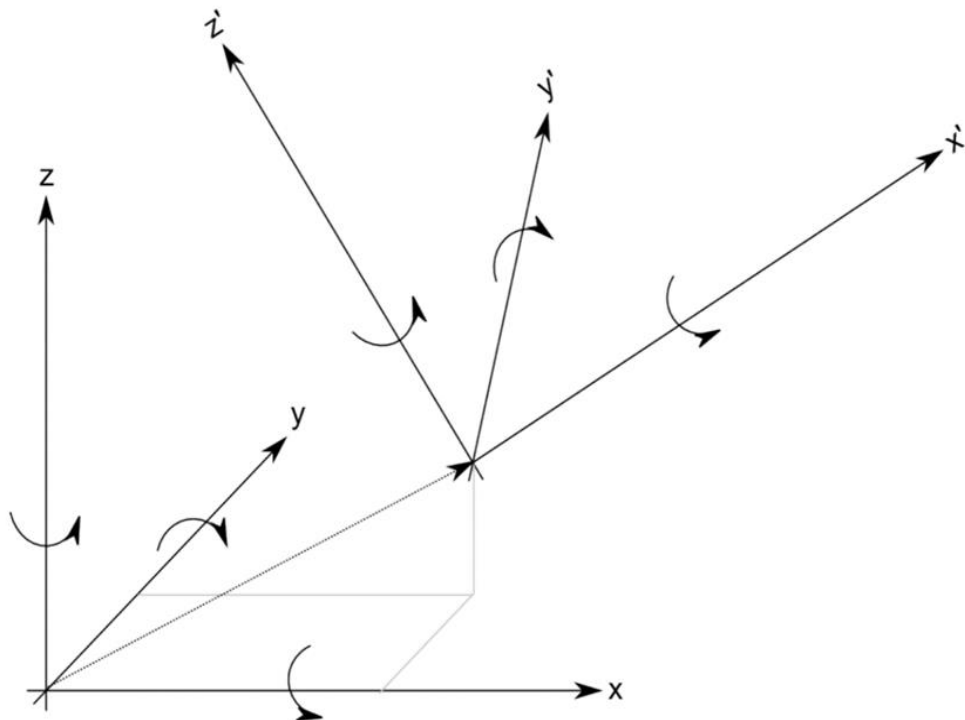


Fig 4.1: Two nested coordinate systems (frames of reference).

Depending on its degrees-of-freedom, that is the number of independent translations and rotations a robot can achieve in Cartesian space, it is also customary to ignore components of position and orientation that remain constant. For example, a simple

floor-cleaning robot's pose might be completely defined by its x and y coordinates in a room as well as its orientation, i.e. its rotation around the z-axis.

4.3 Matrix Notation

Given some kind of fixed coordinate system, we can describe the position of a robot's end-effector by a 3x1 position vector. As there can be many coordinate systems defined on a robot and the environment, we identify the coordinate system a point relates to by a preceding super-script, e.g., ${}^A\mathbf{P}$ to indicate that point P is in coordinate system $\{A\}$. Each point consists of three elements ${}^A\mathbf{P} = [p_x p_y p_z]^T$.

As we know, not only the position of the robot is important, but also its orientation. To describe the orientation of a point, we will attach a coordinate system to it. Let $\hat{X}_B$, $\hat{Y}_B$ and $\hat{Z}_B$ be unit vectors that correspond to the principal axes of a coordinate system $\{B\}$. When expressed in coordinate system $\{A\}$, they are denoted ${}^A\hat{X}_B$, ${}^A\hat{Y}_B$ and ${}^A\hat{Z}_B$. In order to express a vector that is given in one coordinate system in another, we need to project each of its components to the unit vectors that span the target coordinate system.

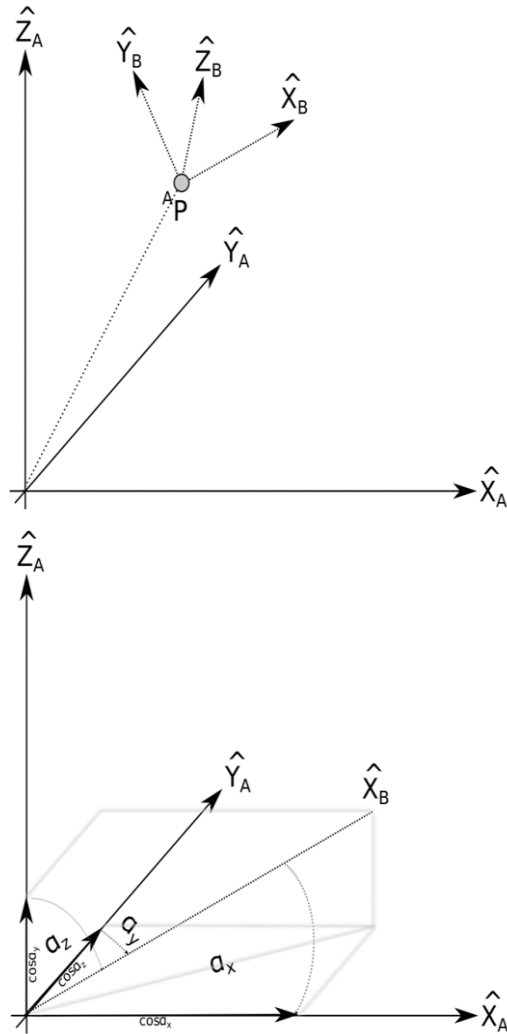


Fig. 4.2 Rotation of frames with some angles

Top: A coordinate system $\{B\}$ with position given by ${}^A P$ and orientation given by $\hat{X}_B$, $\hat{Y}_B$, and $\hat{Z}_B$. Bottom: The projection of the unit vector $\hat{X}_B$ onto the unit vectors that span coordinate system $\{A\}$ after moving $\{B\}$ into the origin of $\{A\}$. As all vectors are unit vectors, $A \cdot B = \|A\| \|B\| \cos \alpha = \cos \alpha$.

which describes $\{B\}$ relative to $\{A\}$. It is important to note that all columns in ${}_B^A R$ are unit vectors, so that the rotation matrix is orthonormal. This is important as this allows us to easily obtain the inverse of ${}_B^A R$ as ${}_B^A R^T$ or ${}_A^B R = {}_B^A R^T$.

where the rotation matrix is nothing but the identity as both points already is in the same coordinate system.

We have now established how to express the orientation of a coordinate system using a rotation matrix. Usually, coordinate systems don't lie on top of each other, but are also displaced from each other.

to describe the coordinate frame $\{B\}$ with respect to $\{A\}$ using a vector ${}^A P$ and a rotation matrix ${}_B^A R$. Robots usually have many such frames defined along their bodies.

Mapping from one frame to another:

Having introduced the concept of frames, we need the ability to map coordinates in one frame to coordinates in another frame. For example, let's consider frame $\{B\}$ having the same orientation as frame $\{A\}$ and sitting at location ${}^A P$ in space.

Adding two vectors that are in different reference frames, i.e., ${}^B Q + {}^A P$, is only possible if both of them have the same orientation. Using this notation, we can see that leading subscripts cancel the leading superscripts of the following vector/rotation matrix. Whereas we have now a solution to transfer a point from one frame of reference to another by combining a rotation and a translation, it would be more appealing to write something like that:

To do this, we need to introduce a 4x1 position vector such that.

$$[{}^A Q] = \left[\begin{array}{c|c} {}_B^A R & {}^A P \\ \hline 0 & 1 \end{array} \right] \left[\begin{array}{c} {}^B Q \\ 1 \end{array} \right]$$

and ${}_B^A T$ is a 4x4 matrix. Note that the added '1's and $[0001]$ do not affect the other entries in the matrix during matrix multiplication. A 4x4 matrix of this form is called a *homogenous transform*.

The inverse of an homogeneous transform can be constructed by inverting rotation and translation part independently, leading to

$$\left[\begin{array}{c|c} {}_B^A R & {}^A P \\ \hline 0 & 1 \end{array} \right]^{-1} = \left[\begin{array}{c|c} {}_B^A R^T & -{}_B^A R^T {}^A P \\ \hline 0 & 1 \end{array} \right]$$

We have now established a convenient notation to convert points from one coordinate system to another. There are many possible ways this can be done, how rotation can be represented (see below), but all can be converted from one into the other.

Transformation arithmetic:

Transformations can be combined: consider for example an arm with two links, reference frame {A} at the base, {B} at its first joint, and {C} at its end-effector. to convert a point in the reference frame of the end-effector to that of its base. As this works for rotation and translation operators independently, we can construct ${}^A_C T$ as

$${}^A_C T = \left[\begin{array}{ccc|c} {}^A_B R {}^B_C R & {}^A_B R {}^B_C P_C + {}^A P_B & 0 & 1 \end{array} \right]$$

where ${}^A P_B$ and ${}^B P_C$ are the translations from {A} to {B} and from {B} to {C}, respectively.

Other representations for orientation:

So far, we have represented orientation by a 3x3 matrix whose column vectors are orthogonal unit vectors describing the orientation of a coordinate system. Orientation is therefore represented with nine different values. We chose this representation mainly because it is the most intuitive to explain and is derived from simple geometry.

In fact, three values are sufficient to describe orientation. This becomes clear when considering that orthogonality (dot product of all columns is zero) and vector length (each vector must have length 1) impose six constraints on the nine values in the rotation matrix. Indeed, an orientation can be represented about a rotation by certain angles around the x, the y, and the z-axis of the reference coordinate system. This is known as the X – Y – Z fixed angle notation. Mathematically, this can be represented by a rotation matrix of the form

$${}^A_B R_{XYZ(\gamma, \beta, \alpha)} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}$$

4.4 D-H Parameters

Now, a robot manipulator can be considered as a set of coordinate frames at each joint. The homogenous transformation matrix can represent the relationship between base and end-effector at any pose.

Consider a robot with ‘n’ number of coordinate frames linked to each other.

Now consider ith frame as {i} and position of a point with respect to the frame as ${}^i P$.

The relationships between the frames are as follows:

$${}^1 P = {}^1 R_2 {}^2 P$$

$${}^2 P = {}^2 R_3 {}^3 P \quad . . .$$

$${}^{n-1}P = {}^{n-1}R_n {}^nP$$

The homogenous transformation matrix between base frame and end-effector is (${}^1P = {}^1R_2 {}^2R_3 \dots {}^{n-1}R_n {}^nP$)

D-H Parameters of the Left Arm of the Robot:

Links	Link length (a)	Link twist (α)	Offset Distance (d)	Angle (θ)
Link 0	0	0	0	0
Link 1	183.211mm	-90	0	variable
Link 2	193mm	0	0	variable

Table 4.1 D-H Parameters of the Left Arm of the Robot

D-H Parameters of the Right Arm of the Robot:

Links	Link length (a)	Link twist (α)	Offset Distance (d)	Angle (θ)
Link 0	0	0	0	0
Link 1	74.081mm	-90	0	variable
Link 2	80mm	0	0	variable
Link 3	112mm	0	0	variable

Table 4.2 D-H Parameters of the Right Arm of the Robot

$${}^{i-1}T_i = \begin{pmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The Homogeneous transformation Matrix for the left arm by using the above D-H parameters can be calculated by

$${}^0P_2 = ({}^0T_1 {}^1T_2 {}^2T_3)$$

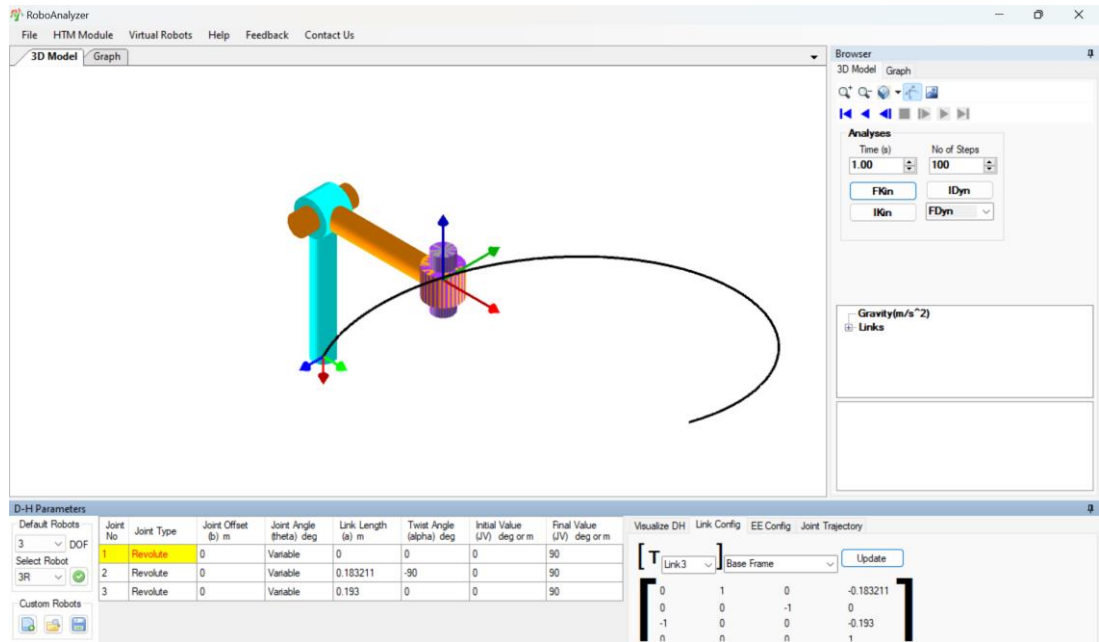


Fig. 4.5 Roboanalyzer software forward kinematic analysis for left arm

Similarly, the D-H parameters for right arm of the robot:

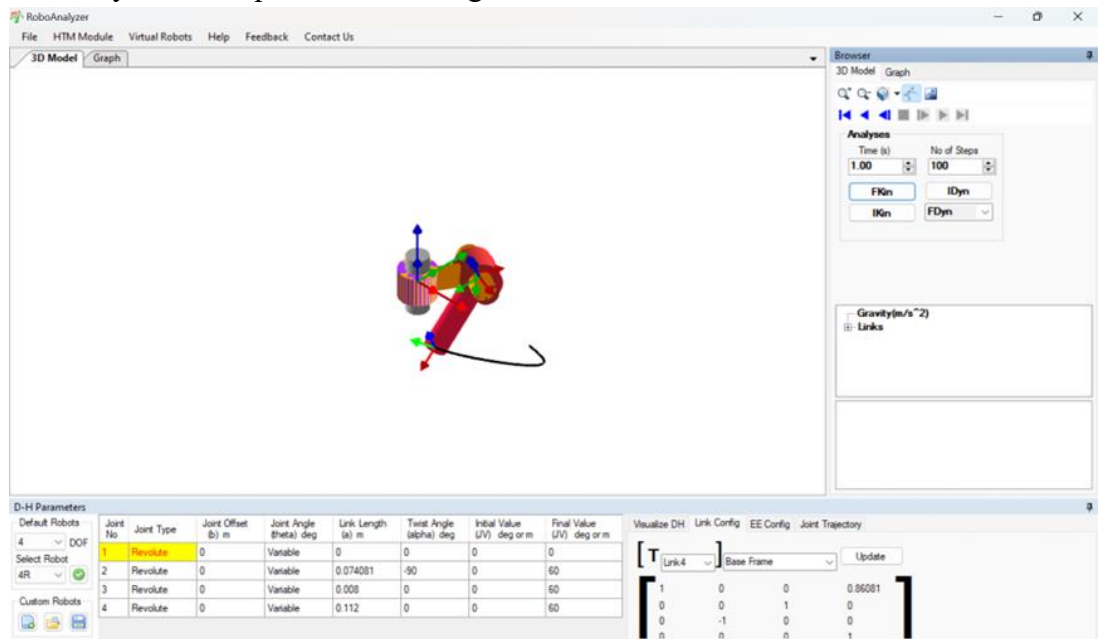


Fig. 4.6 Roboanalyzer software forward kinematic analysis for right arm

4.5 Inverse Kinematic Analysis

Inverse kinematics (IK) is a crucial problem in robotics, especially for manipulators. It involves determining the joint configurations of a robot to achieve a desired end-effector position and orientation. Several methods exist for solving inverse kinematics problems, each with its advantages and disadvantages. Here are some common methods and a brief comparison:

Analytical Methods:

Closed-form solutions: Some manipulators have simple geometries that allow for closed-form solutions. These solutions directly compute joint angles from end-effector position and orientation using mathematical formulas derived from geometric relationships.

Geometric methods: These methods use geometric properties of the manipulator to derive inverse kinematics equations. Examples include geometric interpretation of the manipulator's kinematic chain, such as the law of cosines and the law of sines.

Comparison: Analytical methods provide exact solutions when available, leading to precise and efficient computations. However, closed-form solutions are often limited to specific robot geometries and may not be applicable to complex manipulators.

Numerical Methods:

Iterative methods: Iterative techniques, such as the Jacobian-based methods (e.g., Jacobian transpose, Jacobian pseudoinverse), iteratively adjust joint configurations towards the desired end-effector pose. These methods use the manipulator's Jacobian matrix to map changes in joint space to changes in Cartesian space.

Optimization-based methods: These methods formulate the inverse kinematics problem as an optimization problem, where the objective function minimizes the difference between the desired and actual end-effector pose. Techniques like gradient descent, least squares optimization, or stochastic optimization can be employed.

Comparison:

Numerical methods offer more flexibility and generality compared to analytical methods, as they can handle arbitrary robot geometries and kinematic constraints. However, they may suffer from convergence issues, local minima, and computational overhead, especially for high-dimensional systems.

The inverse kinematics of the robot is done using analytical method for both the Arms:

The Left Arm

The workspace of the left arm is a spherical surface, it can only reach points that are present on the surface of the sphere. The arm is actuated by a prismatic joint and 2 revolute joints leading to a PRR configuration the prismatic joint is along the x-axis. For a point to lie on the surface of the sphere, it should satisfy the sphere locus equation which is.

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

Where x, y, z are the coordinates of the point and a, b, c are coordinates of the center of the sphere which is at joint 3. Now, as the robot translates only in X-axis and c

are fixed r is also fixed which is the length of link 3 now, we need to find the unknown which is a . $a = x \pm \sqrt{(L_3^2 - c)}$

where, $a = x$ -displacement of the prismatic joint

$$c = (y-b)^2 + (z-c)^2$$

Now, that the sphere center (a, b, c) is known we can formulate a transformation matrix between base frame and the sphere's center.

And the matrix is as follows:

$$\begin{pmatrix} 1 & 0 & 0 & -a \\ 0 & 1 & 0 & -b \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Now from the transformation matrix, we get,

$$X_2 = X_1 - a$$

$$Y_2 = Y_1 - b$$

$$Z_2 = Z_1 - c$$

Now the above values form the position vector for the point from the sphere's center with a magnitude of the sphere's radius or the length of link 3.

Now, θ_2 is the angle that the vector forms with z -axis. It can be found by the direction cosines of the vector.

$$\theta_2 = \cos^{-1}(Z_2/L_3) \text{ And, } \theta_1 = \tan^{-1}(Y_2/X_2)$$

$$\text{Therefore, } a = x \pm \sqrt{(L_3^2 - c)}, \theta_2 = \cos^{-1}(Z_2/L_3), \theta_1 = \tan^{-1}(Y_2/X_2)$$

The Right Arm:

The right arm consists of a prismatic joint along X -axis and 3 revolute joints of which the last two are parallel.

For inverse kinematics of the Right arm, it is considered as a 2R planar robot on the x - z plane which is rotated about the Z -axis by the other joint.

This robot has 4 DOF of which 3 are positional and 1 is orientational.

Let x, y, z and θ be the degrees of freedom.

θ_1 is the angle that provides the orientational DOF so $\theta_1 = \theta$.

Now, as θ_1 is known, the x - z plane near joint-3 rotates by that angle.

The transformation matrix for this new coordinate frame with respect to the base is:

$$\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & dx \\ \sin \theta_1 & \cos \theta_1 & 0 & dy \\ 0 & 0 & 0 & dz \\ 0 & 0 & 0 & 1 \end{pmatrix} = {}^1R_2$$

$$\text{Now, } {}^1P = {}^1R_2 {}^2P$$

From the above equation,

We get,

$$x = \cos\theta_1 x_2 + dx$$

$y = \sin\theta_1 x_2 + dy$ as dy , x , y , θ_1 are known,

$$dx = x - (y - dy) \cot\theta_1$$

That is the displacement of prismatic joint.

Now the 2 revolute planar joints on the x-z plane have to move the end effector to (x_2, z_2) .

A robotic arm shown in Figure, is a form of mechanical arm, often programmable, that can either constitute the entire mechanism or be a component within a more intricate robot. These arms comprise links connected by joints that facilitate rotational or linear motion. The links collectively form a kinematic chain, with the chain's end point termed the end effector. This effector, situated at the arm's extremity, is designed for interaction with the surroundings. Kinematic analysis pertains to the relationships among the manipulator's link positions, velocities, and accelerations. It's categorized into direct kinematics and inverse kinematics. In forward kinematics, provided the length of each link and joint angle, the objective is to compute any point's position within the robot's workspace. In inverse kinematics, given the link lengths and a point's position, the aim is to determine the joint angles. The project focuses on a 2R robotic arm with two rotational links operating in a 2D plane, each with a single degree of freedom. Based on the dimensions shown in Figure 1, and using trigonometry:

$$X_0 = Z_0 = 0 \dots\dots\dots (1)$$

$$X_1 = L_1 C_2 \dots\dots\dots (2)$$

$$Z_1 = L_1 S_2 \dots\dots\dots (3)$$

$$X_2 = X_1 + L_2 C_3 \dots\dots\dots (4)$$

$$Z_2 = Z_1 + L_2 S_3 \dots\dots\dots (5)$$

The given equations depict the forward kinematics expressions for a two-link robotic arm. Using trigonometric principles, they ascertain the x and y coordinates of both the arm's joints and its end effector by considering the angles of its joints (θ_1 and θ_2) and the lengths of its links (L_1 and L_2). The starting position is established at the origin, and subsequent positions are computed by adding the horizontal and vertical parts of the link lengths, modified by the respective angles. These equations establish a clear connection between the angles of the joints and the resulting positions of the different elements of the robotic arm.

$$P_x = L_1 C_2 + L_2 C(\theta_2 + \theta_3)$$

$$P_z = L_1 S_2 + L_2 S(\theta_2 + \theta_3)$$

For Inverse kinematics squaring and adding

$$(P_x)^2 = L_1^2 C_2^2 + L_2^2 (C(\theta_2 + \theta_3))^2$$

$$(P_z)^2 = L_1^2 S_2^2 + L_2^2 (S(\theta_2 + \theta_3))^2$$

We get,

$$(P_x)^2 + (P_z)^2 = L_1^2 (S\theta_2^2 + C\theta_2^2) + L_2^2 (S_{23}^2 + C_{23}^2) + 2L_1L_2(C_2C_{23} + S_2S_{23})$$

$$P_x^2 + P_z^2 = L_1^2 + L_2^2 + 2L_1L_2C\theta_3$$

$$C\theta_3 = \frac{P_x^2 + P_z^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$\theta_3 = \text{ArcCos} \left(\frac{P_x^2 + P_z^2 - L_1^2 - L_2^2}{2L_1L_2} \right)$$

$$S\theta_3 = \pm \sqrt{1 - (C\theta_3)^2}$$

$$\theta_3 = \text{aTan2}(S\theta_3, C\theta_3)$$

To find θ_2 :

$$P_x = L_1C_2 + L_2C_{23}$$

$$P_z = L_1S_2 + L_2S_{23}$$

$$P_x = (L_1 + L_2C_3)C_2 + (-L_2S_3)S_2$$

$$P_z = (L_2S_3)C_2 + (L_1 + L_2S_3)S_2$$

Squaring we get,

$$(P_x)^2 + (P_z)^2 = (L_1 + L_2C_3)^2 + (-L_2S_3)^2$$

To get θ_2 , by gramer's rule

$$\begin{pmatrix} P_x \\ P_z \end{pmatrix} = \begin{pmatrix} L_1 + L_2C_3 & -L_2S_3 \\ L_2S_3 & L_1 + L_2C_3 \end{pmatrix}$$

$$(\Delta)C_2 = \begin{pmatrix} P_x & -L_2S_3 \\ P_z & L_1 + L_2C_3 \end{pmatrix}$$

$$C_2 = \frac{P_x(L_1 + L_2C_3) + P_z(L_2S_3)}{(P_x)^2 + (P_z)^2}$$

$$\Delta S_2 = \begin{bmatrix} l1 + l2c3 & Px \\ l2s3 & Py \end{bmatrix}$$

$$S_2 = \frac{\Delta S_2}{2} = \frac{Py(l1 + l2c2) + Px(l2s2)}{Px^2 + Py^2}$$

$$\theta_2 = \text{aTan2}(s3, c3)$$

CHAPTER 05

SOFTWARE

5.1 Introduction

The development of a digital robotic simulator involves the different functionalities of the simulator implemented by different software and technologies.

Some of the key necessities for this are:

- 3D model that can be animated to simulate the motion of the robot.
- Database to store and access data regarding various aspects of the robot-like D-H parameters, motion planning, work volumes, etc...
- Webpage to serve as a user interface and web server to host the webpage and database.
- Integration of the above functionalities into a single piece of software.

As the software was being created from scratch, the different components of the software had to be surveyed and the software best suited for the purpose was chosen.

Digital simulators/ digital twin: A digital twin is a digital model of an intended or actual real-world physical product, system, or process (a physical twin) that serves as the effectively indistinguishable digital counterpart of it for practical purposes, such as simulation, integration, testing, monitoring, and maintenance. The digital twin has been intended from its initial introduction to be the underlying premise for Product Lifecycle Management and exists throughout the entire lifecycle (create, build, operate/support, and dispose) of the physical entity it represents. Since information is granular, the digital twin representation is determined by the value-based use cases it is created to implement. The digital twin can and does often exist before there is a physical entity, as for example with virtual prototyping. The use of a digital twin in the creation phase allows the intended entity's entire lifecycle to be modeled and simulated. A digital twin of an existing entity may be used in real-time and regularly synchronized with the corresponding physical system.

An example of digital twins is the use of 3D modeling to create digital companions for physical objects. It can be used to view the status of the actual physical object, which provides a way to project physical objects into the digital world. For example, when sensors collect data from a connected device, the sensor data can be used to update a "digital twin" copy of the device's state in real time. The term "device shadow" is also used for the concept of a digital twin. The digital twin is meant to be an up-to-date and accurate copy of the physical object's properties and states, including shape, position, gesture, status and motion.

A digital twin also can be used for monitoring, diagnostics, and prognostics to optimize asset performance and utilization. In this field, sensory data can be combined with historical data, human expertise and fleet and simulation learning to improve the outcome of prognostics. Therefore, complex prognostics and intelligent

maintenance system platforms can use digital twins in finding the root cause of issues and improve productivity.

Digital twins of autonomous vehicles and their sensor suite embedded in a traffic and environment simulation have also been proposed to overcome the significant development, testing and validation challenges for the automotive application, in particular when the related algorithms are based on artificial intelligence approaches that require extensive training data and validation data sets.

5.2 Choosing the tech stack and software

The chosen components for this purpose are:

- CATIA for mechanical design of robots.
- Blender 3D to animate the model and process the kinematics.
- HTML, CSS to create webpages and serve as a UI.
- PHP as the backend of the software.
- MySQL as the database.
- Three.js to display the 3D model and animations in the software.
- The functionalities of the above software's were integrated to create the digital simulator.

Integration with other industrial systems in the era of Industry 4.0.

5.3 Blender 3D

Blender is a free and open-source 3D creation suite. With Blender, you can create 3D visualizations such as still images, 3D animations.

Key Features :

- Blender is a fully integrated 3D content creation suite, offering a broad range of essential tools, including Modeling, Rendering, Animation & Rigging, Video Editing, VFX, Compositing, Texturing, and many types of Simulations.
- It is cross platform, with an OpenGL GUI that is uniform on all major platforms (and customizable with Python scripts).
- It has a high-quality 3D architecture, enabling fast and efficient creation workflow.
- It boasts active community support. See blender.org/community for an extensive list of sites.
- It has a small executable, which is optionally portable.

Blender offers a variety of different binary packages to choose from depending on their level of stability. Each package has the tradeoff of newest feature versus stability. The package that is right for you depends on your requirements for those two. A studio for example might want to have long-term support, while a hobbyist may want newer features, while others may just want to test upcoming features. Each package described below has something just right for everyone.

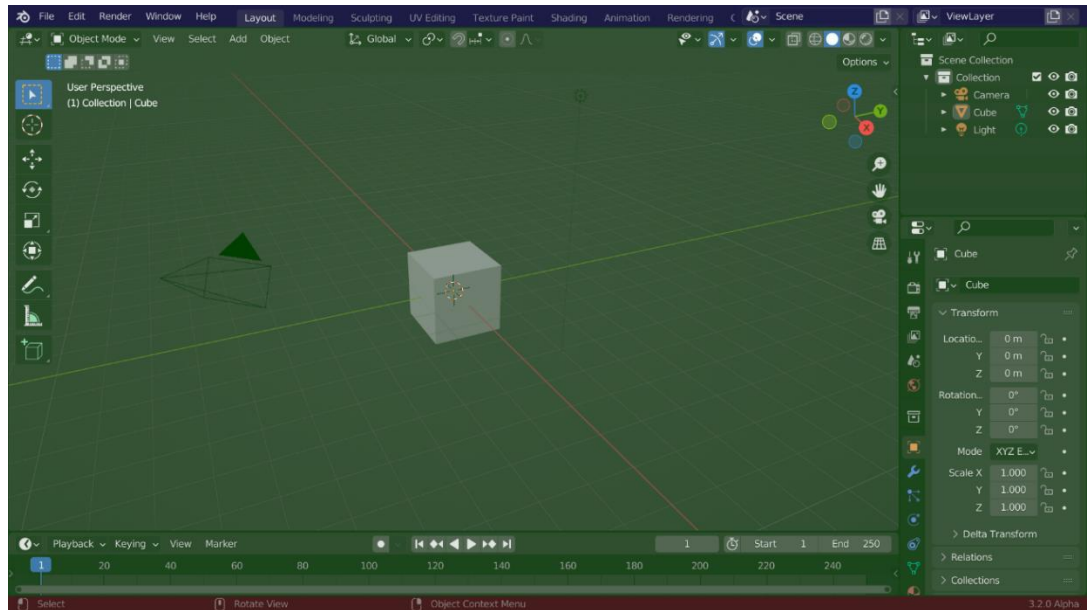


Fig 5.1 Blender Home Page

Blender Python API features:

- Edit any data the user interface can (Scenes, Meshes, Particles etc.).
- Modify user preferences, keymaps and themes.
- Run tools with own settings.
- Create user interface elements such as menus, headers, and panels.
- Create new tools.
- Create interactive tools.
- Create new rendering engines that integrate with Blender.
- Subscribe to changes to data and it's properties.
- Define new settings in existing Blender data.
- Draw in the 3D Viewport using Python.

Running Scripts

The two most common ways to execute Python scripts are using the built-in text editor or entering commands in the Python console. Both the Text Editor and Python Console are space types you can select from the header. Rather than manually configuring your spaces for Python development, you can use the Scripting workspace accessible from the Topbar tabs.

From the text editor you can open .py files or paste then from the clipboard, then test using Run Script. The Python Console is typically used for typing in snippets and for testing to get immediate feedback, but can also have entire scripts pasted into it. Scripts can also run from the command line with Blender but to learn scripting in Blender this isn't essential.

Data Access

Accessing Data-Blocks

You can access Blender's data with the Python API in the same way as the animation system or user interface; this implies that any setting that can be changed via a button

can also be changed with Python. Accessing data from the currently loaded blend-file is done with the module `bpy.data`. It gives access to library data, for example:

```
>>> bpy.data.objects
<bpy_collection, BlendDataObjects>
>>> bpy.data.scenes
<bpy_collection, BlendDataScenes>
>>> bpy.data.materials
<bpy_collection[1], BlendDataMaterials>
```

Accessing Collections

You will notice that an index as well as a string can be used to access members of the collection. Unlike Python dictionaries, both methods are available; however, the index of a member may change while running Blender.

```
>>> list(bpy.data.objects)
[bpy.data.objects["Cube"], bpy.data.objects["Plane"]]
>>> bpy.data.objects['Cube']
bpy.data.objects["Cube"]
>>> bpy.data.objects[0]
bpy.data.objects["Cube"]
```

Accessing Attributes

Once you have a data-block, such as a material, object, collection, etc., its attributes can be accessed much like you would change a setting using the graphical interface. In fact, the tooltip for each button also displays the Python attribute which can help in finding what settings to change in a script.

```
>>> bpy.data.objects[0].name 'Camera'
>>> bpy.data.scenes["Scene"] bpy.data.scenes['Scene']
>>> bpy.data.materials.new("MyMaterial") bpy.data.materials['MyMaterial']
```

For testing what data to access it's useful to use the Python Console, which is its own space type. This supports auto-complete, giving you a fast way to explore the data in your file.

Example of a data path that can be quickly found via the console:

```
>>> bpy.data.scenes[0].render.resolution_percentage100
>>> bpy.data.scenes[0].objects["Torus"].data.vertices[0].co.x1.0
```

Data Creation/Removal

When you are familiar with other Python APIs you may be surprised that new data-blocks in the `bpy` API cannot be created by calling the class:

```
>>> bpy.types.Mesh()
```

Traceback (most recent call last):

```
File "<blender_console>", line 1, in <module>
```

```
TypeError: bpy_struct.__new__(type): expected a single argument
```

This is an intentional part of the API design. The Blender Python API can't create Blender data that exists outside the main Blender database (accessed through `bpy.data`), because this data is managed by Blender (save, load, undo, append, etc).

Data is added and removed via methods on the collections in `bpy.data`, e.g:

```
>>> mesh = bpy.data.meshes.new(name="MyMesh")
>>> print(mesh)
<bpy_struct, Mesh("MyMesh.001")>
>>> bpy.data.meshes.remove(mesh)
```

Custom Properties

Python can access properties on any data-block that has an ID (data that can be linked in and accessed from `bpy.data`). When assigning a property, you can pick your own names, these will be created when needed or overwritten if they already exist.

This data is saved with the blend-file and copied with objects, for example:

```
bpy.context.object["MyOwnProperty"] = 42
if "SomeProp" in bpy.context.object:
    print("Property found")
# Use the get function like a Python dictionary
# which can have a fallback value.
value = bpy.data.scenes["Scene"].get("test_prop", "fallback value")
# dictionaries can be assigned as long as they only use basic types.
collection = bpy.data.collections.new("MyTestCollection")
collection["MySettings"] = {"foo": 10, "bar": "spam", "baz": {}}
del collection["MySettings"]
```

Note that these properties can only be assigned basic Python types:

- int, float, string
- array of ints or floats
- dictionary (only string keys are supported, values must be basic types too)

These properties are valid outside of Python. They can be animated by curves or used in driver paths.

Context

While it's useful to be able to access data directly by name or as a list, it's more common to operate on the user's selection. The context is always available from `bpy.context` and can be used to get the active object, scene, tool settings along with many other attributes.

Some common use cases are:

```
>>> bpy.context.object
>>> bpy.context.selected_objects
>>> bpy.context.visible_bones
```

Note that the context is read-only, which means that these values cannot be modified directly. But they can be changed by running API functions or by using the data API. So, `bpy.context.active_object = obj` will raise an error. But `bpy.context.view_layer.objects.active = obj` works as expected.

The context attributes change depending on where they are accessed. The 3D Viewport has different context members than the Python Console, so take care when accessing context attributes that the user state is known.

See `bpy.context` API reference.

Operators (Tools)

Operators are tools generally accessed by the user from buttons, menu items or key shortcuts. From the user perspective they are a tool but Python can run these with its own settings through the `bpy.ops` module.

Examples:

```
>>> bpy.ops.mesh.flip_normals() {'FINISHED'}
>>> bpy.ops.mesh.hide(unselected=False) {'FINISHED'}
>>> bpy.ops.object.transform_apply() {'FINISHED'}
```

Many operators have a “poll” function which checks if the cursor is in a valid area or if the object is in the correct mode (Edit Mode, Weight Paint Mode, etc). When an operator’s poll function fails within Python, an exception is raised.

For example, calling `bpy.ops.view3d.render_border()` from the console raises the following error:

RuntimeError: Operator `bpy.ops.view3d.render_border.poll()` failed, context is incorrect

In this case the context must be the 3D Viewport with an active camera.

To avoid using try-except clauses wherever operators are called, you can call the operator’s own `poll()` function to check if it can run the operator in the current context.

5.4 LAMPP Stack in web-development

Linux-Apache-Mysql-PHP stack

Linux:

Linux is a family of open-source Unix-like operating systems based on the Linux kernel, an operating system kernel first released on September 17, 1991, by Linus Torvalds. Linux is typically packaged as a Linux distribution (distro), which includes the kernel and supporting system software and libraries, many of which are provided by the GNU Project. Many Linux distributions use the word "Linux" in their name, but the Free Software Foundation uses and recommends the name "GNU/Linux" to emphasize the use and importance of GNU software in many distributions, causing some controversy.

Popular Linux distributions include Debian, Fedora Linux, Arch Linux, and Ubuntu. Commercial distributions include Red Hat Enterprise Linux and SUSE Linux Enterprise. Desktop Linux distributions include a windowing system such as X11 or Wayland and a desktop environment such as GNOME or KDE Plasma. Distributions intended for servers may not have a graphical user interface at all, or include a solution stack such as LAMP. Because Linux is freely redistributable, anyone may create a distribution for any purpose.

Linux was originally developed for personal computers based on the Intel x86 architecture but has since been ported to more platforms than any other operating system. Because of the dominance of Linux-based Android on smartphones, Linux, including Android, has the largest installed base of all general-purpose operating systems as of May 2022. Although Linux is, as of November 2022, used by only around 2.6 percent of desktop computers the Chromebook, which runs the Linux kernel-based ChromeOS, dominates the US K–12 education market and represents nearly 20 percent of sub-\$300 notebook sales in the US.[27] Linux is the leading operating system on servers (over 96.4% of the top one million web servers' operating systems are Linux), leads other big iron systems such as mainframe computers, and is used on all of the world's 500 fastest supercomputers (as of November 2017, having gradually displaced all competitors).

Apache Server:

The Apache HTTP Server is a free and open-source cross-platform web server software, released under the terms of Apache License 2.0. It is developed and maintained by a community of developers under the auspices of the Apache Software Foundation.

The vast majority of Apache HTTP Server instances run on a Linux distribution, but current versions also run on Microsoft Windows, OpenVMS, and a wide variety of Unix-like systems. Past versions also ran on NetWare, OS/2 and other operating systems, including ports to mainframes.

My SQL:

MySQL is an open-source relational database management system (RDBMS). Its name is a combination of "My", the name of co-founder Michael Widenius's daughter My, and "SQL", the acronym for Structured Query Language. A relational database organizes data into one or more data tables in which data may be related to each other; these relations help structure the data. SQL is a language that programmers use to create, modify and extract data from the relational database, as well as control user access to the database. In addition to relational databases and SQL, an RDBMS like MySQL works with an operating system to implement a relational database in a computer's storage system, manages users, allows for network access and facilitates testing database integrity and creation of backups.

MySQL is free and open-source software under the terms of the GNU General Public License, and is also available under a variety of proprietary licenses. MySQL was owned and sponsored by the Swedish company MySQL AB, which was bought by Sun Microsystems (now Oracle Corporation). In 2010, when Oracle acquired Sun, Widenius forked the open-source MySQL project to create MariaDB.

MySQL has stand-alone clients that allow users to interact directly with a MySQL database using SQL, but more often, MySQL is used with other programs to implement applications that need relational database capability. MySQL is a component of the LAMP web application software stack (and others), which is an acronym for Linux, Apache, MySQL, Perl/PHP/Python.

PHP :

PHP is a general-purpose scripting language geared towards web development. It was originally created by Danish-Canadian programmer Rasmus Lerdorf in 1993 and released in 1995. The PHP reference implementation is now produced by the PHP Group. PHP was originally an abbreviation of Personal Home Page, but it now stands for the recursive initialism PHP: Hypertext Preprocessor.

PHP code is usually processed on a web server by a PHP interpreter implemented as a module, a daemon or a Common Gateway Interface (CGI) executable. On a web server, the result of the interpreted and executed PHP code—which may be any type of data, such as generated HTML or binary image data—would form the whole or part of an HTTP response. Various web template systems, web content management systems, and web frameworks exist that can be employed to orchestrate or facilitate the generation of that response. Additionally, PHP can be used for many programming tasks outside the web context, such as standalone graphical applications and drone control. PHP code can also be directly executed from the command line

5.5 Using Three.js to display 3D Model

Three.js is a cross-browser JavaScript library and application programming interface (API) used to create and display animated 3D computer graphics in a web browser using WebGL. The source code is hosted in a repository on GitHub.

Three.js allows the creation of graphical processing unit (GPU)-accelerated 3D animations using the JavaScript language as part of a website without relying on proprietary browser plugins. This is possible due to the advent of WebGL, a low-level graphics API created specifically for the web.

High-level libraries such as Three.js or GLGE, Scene.js, PhiloGL, and many more make it possible to author complex 3D computer animations for display in the browser without the effort required for a traditional standalone application or a plugin.

Three.js was first released by Ricardo Cabello on GitHub in April 2010. The origins of the library can be traced back to his involvement with the demoscene in the early 2000s. The code was originally developed in the ActionScript language used by Adobe Flash, later being ported to JavaScript in 2009. In Cabello's mind, there were two strong points that justified the shift away from ActionScript: Firstly, JavaScript provided greater platform independence. Secondly, applications written in JavaScript would not need to be compiled by the developer beforehand, unlike Flash applications. Additional contributions by Cabello include API design, CanvasRenderer, SVGRenderer, and being responsible for merging the commits by the various contributors into the project. With the advent of WebGL, Paul Brunt was able to implement the new rendering technology quite easily as Three.js was designed with the rendering code as a module rather than in the core itself. Branislav Uličný, an early contributor, started with Three.js in 2010 after having posted a number of

WebGL demos on his own site. He wanted WebGL renderer capabilities in Three.js to exceed those of CanvasRenderer or SVGRenderer. His major contributions generally involve materials, shaders, and post-processing.

Soon after the introduction of WebGL 1.0 on Firefox 4 in March 2011, Joshua Koo came on board. He built his first Three.js demo for 3D text in September 2011. His contributions frequently relate to geometry generation. Starting from version 118, Three.js uses WebGL 2.0 by default. Older version of the standard is still available via WebGL1Renderer class. Three.js has over 1700 contributors on GitHub.

5.6 Integrating the different components

The coding of the software is shown below:

Front-end:

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta http-equiv='cache-control' content='no-cache'>
  <meta http-equiv='expires' content='0'>
  <meta http-equiv='pragma' content='no-cache'>
  <script type="module" src="https://ajax.googleapis.com/ajax/libs/model-viewer/3.3.0/model-viewer.min.js"></script>
  <title>Set angles</title>
</head>
<body>
  <model-viewer src="scene.glb" autoplay animation-name="left2Action" ar ar-
modes="webxr scene-viewer quick-look" camera-controls poster="poster.webp"
shadow-intensity="1" style="height: 100%;width: 50%;background-color:
black;border-radius: 10%;">
    <button slot="ar-button" id="ar-button">
      View in your space
    </button>
  </model-viewer><br>
  <pre style="position: absolute;left: 51%;top: 25%;"><form action="link.php"
method="post"><h3>
    enter left 1 value: <input type="number" name="left1"><br><br>
    enter left 2 value: <input type="number" name="left2"><br><br>
    enter right 1 value: <input type="number" name="right1"><br><br>
    enter right 2 value: <input type="number" name="right2"><br><br>
    enter right 3 value: <input type="number" name="right3"><br><br>
    enter right 4 value: <input type="number" name="right4"><br><br>
    enter x value: <input type="number" name="x"><br><br>
    enter y value: <input type="number" name="y"><br><br></h3>
```

```

        <button type="submit">Submit</button>
    </form></pre>
    <h3><a href="index2.html">SET WORK VOLUMES</a></h3>
</body>
</html>

```

Back-end :

```

<?php
    // servername => localhost
    // username => root
    // password => empty
    // database name => angles
    $conn = mysqli_connect("localhost", "root", "", "digital twin");
    // Check connection
    if($conn === false){
        die("ERROR: Could not connect. "
            . mysqli_connect_error());
    }
    // Taking all 5 values from the form data(input)
    $left1 = $_REQUEST['left1'];
    $left2 = $_REQUEST['left2'];
    $right1 = $_REQUEST['right1'];
    $right2 = $_REQUEST['right2'];
    $right3 = $_REQUEST['right3'];
    $right4 = $_REQUEST['right4'];
    $x = $_REQUEST['x'];
    $y = $_REQUEST['y'];
    // Performing insert query execution
    $sql = "UPDATE angles SET
left1='$left1',left2='$left2',right1='$right1',right2='$right2',right3='$right3',right4='$
right4',x='$x',y='$y'";
    if(mysqli_query($conn, $sql)){
        echo "<h3>data stored in a database successfully."
            . " Please browse your localhost php my admin"
            . " to view the updated data</h3>";
        echo nl2br("\n$left1\n $left2\n $right1\n $right2\n $right3\n $right4\n $x\n
$y\n ");
    } else{
        echo "ERROR: Hush! Sorry $sql. "
            . mysqli_error($conn);
    }
    // Close connection
    mysqli_close($conn);

```

```

exec("python3 pythoncode.py");
header("Location: index.html");
exit;
?>

```

Python code :

```

import bpy #blender API for python
import math
import mysql.connector as sc
db=sc.connect(
    host="localhost",
    user="root",
    password="",
    database="digital twin")
cursor=db.cursor()
cursor.execute("SELECT left1,left2,right1,right2,right3,right4,x,y FROM angles")
result = cursor.fetchall()
left1 = result[0][0]
left2 = result[0][1]
right1 = result[0][2]
right2 = result[0][3]
right3 = result[0][4]
right4 = result[0][5]
x = result[0][6]
y = result[0][7]
#open blender file
bpy.ops.wm.open_mainfile(filepath="model.blend")
#add keyframes at current location
bpy.data.objects["left1"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["left2"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["right1"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["right2"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["right3"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["right4"].keyframe_insert(data_path="rotation_euler",frame=1)
bpy.data.objects["carriagex"].keyframe_insert(data_path="location",frame=1)
bpy.data.objects["carriagey"].keyframe_insert(data_path="location",frame=1)
#end of add keyframes
#apply rotations by naming objects
bpy.data.objects["left1"].rotation_euler[2] = 1.5708+math.radians(left1)
bpy.data.objects["left2"].rotation_euler[0] = 1.5708+math.radians(left2)
bpy.data.objects["right1"].rotation_euler[1] = 1.5708+math.radians(right1)
bpy.data.objects["right2"].rotation_euler[0] = 0+math.radians(right2)
bpy.data.objects["right3"].rotation_euler[0] = 1.5708+math.radians(right3)

```

```

bpy.data.objects["right4"].rotation_euler[1] = -1.5708+math.radians(right4)
#translations
bpy.data.objects["carriagex"].location.x -= x/100
bpy.data.objects["carriagey"].location.y += y/100
#add keyframes at new location
bpy.data.objects["left1"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["left2"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["right1"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["right2"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["right3"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["right4"].keyframe_insert(data_path="rotation_euler",frame=160)
bpy.data.objects["carriagex"].keyframe_insert(data_path="location",frame=160)
bpy.data.objects["carriagey"].keyframe_insert(data_path="location",frame=160)
#end of add keyframes
#save as gltf
scene = bpy.context.scene
bpy.ops.export_scene.gltf(filepath="scene.glb")
#save blender file
bpy.ops.wm.save_as_mainfile(filepath=bpy.data.filepath
Python code for work-volumes :
import bpy #blender API for python
import math
import mysql.connector as sc
db=sc.connect(
    host="localhost",
    user="root",
    password="",
    database="digital twin")
cursor=db.cursor()
cursor.execute("SELECT `x`,`y`,`z`,`x-dist`,`y-dist`,`z-dist`,`cubes`,`rcube` FROM
`work volume`")
result = cursor.fetchall()
x = result[0][0]
y = result[0][1]
z = result[0][2]
xdist = result[0][3]
ydist = result[0][4]
zdist = result[0][5]
name = result[0][6]
rname = result[0][7]
rname = str(rname)

```

```

#open blender file
bpy.ops.wm.open_mainfile(filepath="model.blend")
#take inputs
#left1=int(input("Enter left 1 angle: "))
#left2=int(input("Enter left 2 angle: "))
#right1=int(input("Enter right 1 angle: "))
#right2=int(input("Enter right 2 angle: "))
#right3=int(input("Enter right 3 angle: "))
#right4=int(input("Enter right 4 angle: "))
#x=int(input("Enter X value:"))
#y=int(input("Enter y value:"))
if x>0 and y>0 and z>0 :
    #add cube at location
    bpy.ops.mesh.primitive_cube_add(location=(xdist+(x/2),ydist+(y/2),zdist+(z/2)))
    #set size of cube
    bpy.context.active_object.dimensions = (x,y,z)
    bpy.context.active_object.name = str(name)
    #delete cube
    if rname != '0' :
        bpy.ops.object.select_all(action='DESELECT')
        bpy.data.objects[rcube].select_set(True)
        bpy.ops.object.delete()
        cursor.execute("UPDATE `work volume` SET `rname`='0' WHERE 1")
    #save as gltf
    scene = bpy.context.scene
    bpy.ops.export_scene.gltf(filepath="scene.glb")
    #save blender file
    bpy.ops.wm.save_as_mainfile(filepath=bpy.data.filepath)

```


CHAPTER 06

RESULT AND DISCUSSION

Designing and implementing a control framework for Industry 4.0 in industrial robotics is a complex endeavour that requires careful consideration of various factors such as connectivity, adaptability, efficiency, and safety. The result and discussion section of such a framework would typically encompass the outcomes of the design process, the performance of the framework in practical applications, and an analysis of its strengths and limitations. Below, I'll outline the key aspects typically covered in this section:

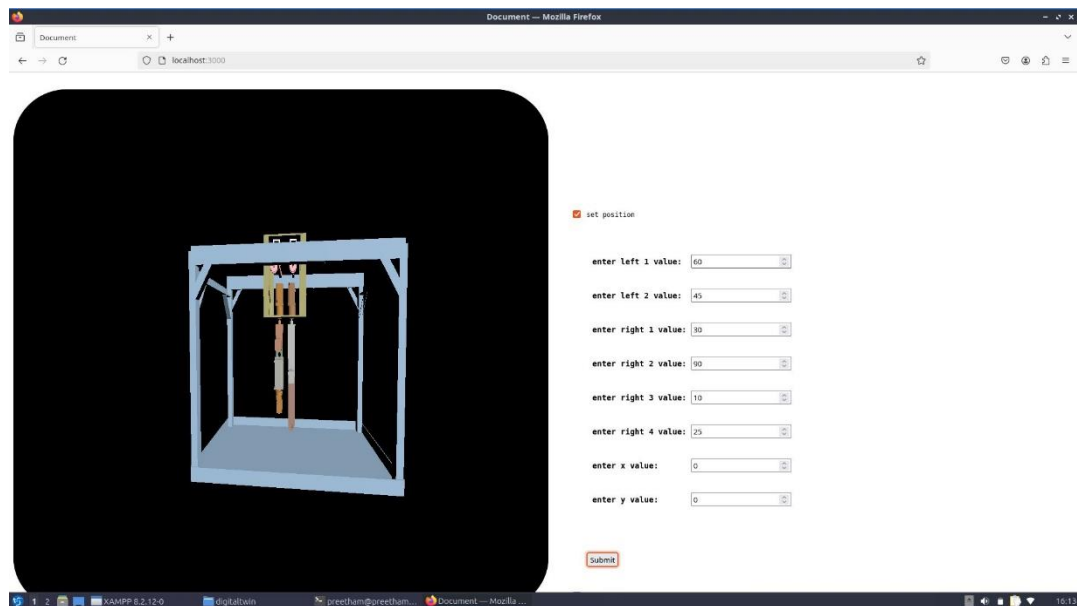


Fig 6.1 Webview to input angles.

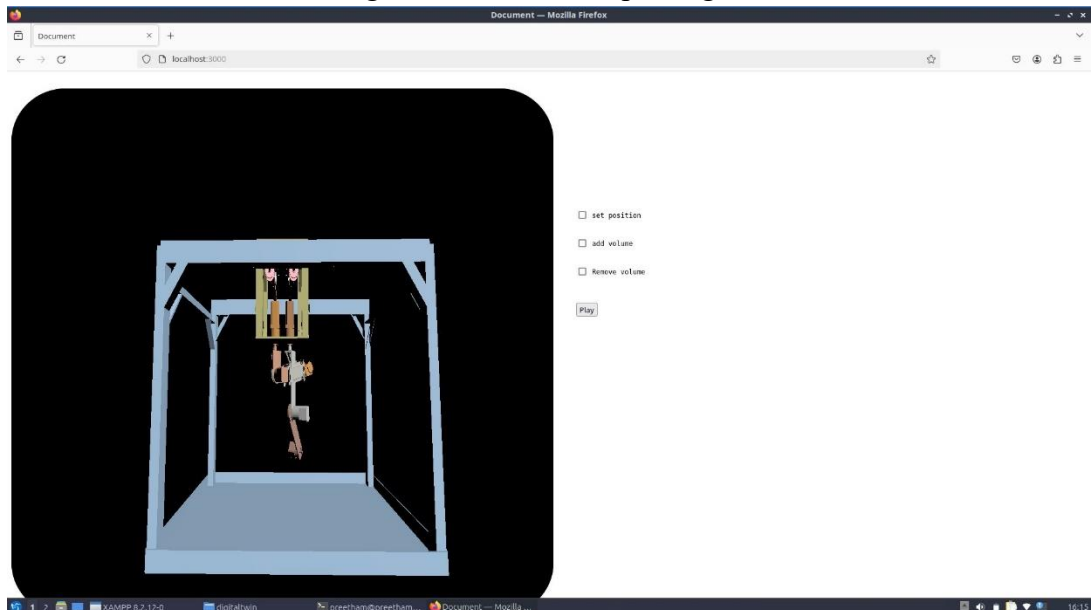


Fig 6.2 result.

Framework Design: Describe the architecture and components of the designed control framework, including hardware (such as sensors, actuators, controllers) and software (such as communication protocols, data processing algorithms, decision-making modules).

Integration with Industry 4.0 Technologies: Discuss how the framework integrates with other Industry 4.0 technologies such as IoT, cloud computing, big data analytics, and artificial intelligence to enable functionalities like predictive maintenance, real-time monitoring, and adaptive control.

Performance Evaluation: Present the results of performance tests conducted on the framework. This could include metrics such as speed, accuracy, reliability, energy efficiency, and adaptability to different manufacturing scenarios.

Case Studies: Provide case studies or examples demonstrating the application of the framework in real-world industrial settings. Highlight specific tasks or processes where the framework has been successfully implemented and its impact on productivity, quality, and cost-effectiveness.

Safety and Compliance: Discuss how the framework ensures safety standards and regulatory compliance in industrial environments. Present any safety mechanisms or protocols implemented within the framework to mitigate risks associated with robotic operations.

Discussion:

Advantages: Discuss the advantages of the designed framework compared to traditional industrial robotic control systems. This may include improved flexibility, scalability, interoperability, and adaptability to dynamic manufacturing environments.

Challenges and Limitations: Address the challenges and limitations encountered during the design and implementation process. This could involve technical challenges, such as interoperability issues with legacy systems, as well as practical limitations, such as high initial investment costs or difficulties in workforce training.

Future Directions: Propose potential areas for future research and development to enhance the capabilities of the framework. This may include exploring advanced control algorithms, integrating emerging technologies, or addressing specific industry needs such as human-robot collaboration or autonomous decision-making.

Impact on Industry 4.0 Adoption: Discuss how the designed framework contributes to the broader adoption of Industry 4.0 principles in manufacturing industries. Analyse its

potential to drive innovation, improve competitiveness, and foster digital transformation across various sectors.

Sustainability and Environmental Impact: Consider the environmental implications of deploying the framework, such as energy consumption, waste reduction, and carbon footprint. Discuss strategies for optimizing resource utilization and minimizing environmental impact in industrial robotic operations.

By presenting comprehensive results and engaging in thoughtful discussions, the result and discussion section provides valuable insights into the design and control framework for Industry 4.0 in industrial robotics, paving the way for further advancements in this field.

CHAPTER 07

FUTURE SCOPE AND WORK

The future scope of work for the project "Design and Control Framework for Industry Robotic 4.0" involves several avenues for further research, development, and implementation. Here are some potential future directions:

- 1) **Advanced Control Algorithms:** Explore and develop more sophisticated control algorithms that leverage artificial intelligence (AI) and machine learning (ML) techniques. This includes reinforcement learning, deep learning, and fuzzy logic-based control strategies to enhance the adaptability, autonomy, and efficiency of robotic systems in Industry 4.0 environments.
- 2) **Human-Robot Collaboration:** Investigate methods for seamless collaboration between human workers and robotic systems in industrial settings. Develop safety protocols, intuitive interfaces, and ergonomic designs to facilitate effective cooperation and task sharing between humans and robots, leading to improved productivity and workplace safety.
- 3) **Interoperability and Standardization:** Address interoperability challenges by promoting the development and adoption of industry-wide standards for communication protocols, data formats, and interfaces. Facilitate seamless integration and compatibility between different robotic systems, manufacturing equipment, and information systems within the industry 4.0 framework.
- 4) **Edge Computing and Edge Robotics:** Explore the potential of edge computing technologies to enhance the capabilities of robotic systems by enabling real-time data processing, decision-making, and control at the edge of the network. Develop edge robotics solutions that minimize latency, reduce reliance on centralized infrastructure, and improve responsiveness in dynamic manufacturing environments.
- 5) **Robotic Swarms and Multi-Agent Systems:** Investigate the use of robotic swarms and multi-agent systems to tackle complex manufacturing tasks collaboratively. Develop coordination algorithms, communication protocols, and distributed control strategies to enable swarms of robots to work together autonomously, adaptively, and efficiently in large-scale production scenarios.
- 6) **Cybersecurity and Resilience:** Address cybersecurity concerns related to interconnected robotic systems in Industry 4.0 environments. Develop robust security measures, encryption techniques, and intrusion detection systems to safeguard sensitive data, prevent unauthorized access, and ensure the resilience of industrial robotic networks against cyber threats and attacks.
- 7) **Lifecycle Management and Maintenance:** Develop comprehensive lifecycle management and predictive maintenance strategies for robotic systems in Industry

4.0. Implement condition monitoring, predictive analytics, and remote diagnostics capabilities to optimize asset performance, minimize downtime, and extend the lifespan of robotic equipment through proactive maintenance and repair.

- 8) **Sustainability and Green Robotics:** Explore strategies for integrating sustainability principles into the design and operation of robotic systems in Industry 4.0. Develop energy-efficient algorithms, recyclable materials, and eco-friendly manufacturing processes to minimize environmental impact, reduce resource consumption, and promote sustainable practices in industrial robotics.

By pursuing these future research directions, the project on designing and controlling frameworks for Industry Robotic 4.0 can contribute to advancing the state-of-the-art in robotic technology, driving innovation, and enabling transformative changes in manufacturing industries towards more efficient, flexible, and sustainable production systems.

CHAPTER 08

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